

ACSD: An Executable Profile for Anonymous Scholarly Releases, Series Lineage, and Cyclic Citations

Anonymous authors

Abstract

Anonymous scholarly disclosure needs a more precise answer than “put a signed PDF in a repository.” A signature identifies a signing key, not necessarily a person; a repository timestamp is not independent time; and mutually citing papers appear circular if every reference is a content hash. This paper presents ACSD, a narrow executable profile for anonymous scholarly releases. A compact approval target binds exact release, governance, and provenance-policy digests before every required pseudonymous key signs it. A separate optional series layer binds stable random *WorkIDs* to completed content-addressed releases, separating a possibly cyclic semantic citation graph from an acyclic commitment graph. The implementation uses standard tagged Ed25519/EdDSA COSE, derives claims from verified capabilities rather than policy labels, and grants external time only for an RFC 3161 response checked against a verifier-supplied signer pin. Evaluation includes 64 fixed and 1,000 generated Python–Node canonical-JSON cases, an independent Node approval verifier, adversarial timestamp tests, a real TSA interoperability run, and 16 Lean theorems over the composition rules. We state the boundary explicitly: none of these mechanisms proves natural-person identity, contribution truth, originality, peer review, or legal nonrepudiation.

1 Introduction

Researchers often want to disclose a manuscript before conventional publication while keeping the author identity unavailable during review. A common informal proposal is to upload an anonymous file, sign it, and rely on repository history. That proposal leaves several distinct questions entangled.

One concrete trigger is the interval around an arXiv submission: a first-time or new-category submission may require endorsement, while moderation and review are separate from scientific peer review. During that interval an author may want a public, content-anonymous evidence package that preserves exact bytes and team assent without pretending to replace arXiv’s process. We use this as an application scenario, not as a claim that arXiv queues have a measured backlog or that ACSD should circumvent its rules.

1. What exact bytes did one or more keys approve?
2. Can a later verifier distinguish a revision, a legitimate branch, and an equivocation?
3. Can a sequence of papers cite one another, including mutual citations, without a circular content-hash dependency?
4. What does a time witness establish, and what does it not establish?

The first question is about key control and integrity. It is not, by itself, a proof that a particular natural person wrote the work. The second and third are data-model and verification questions.

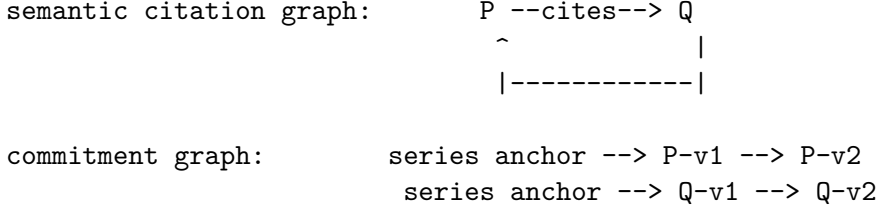


Figure 1: ACSD separates cyclic semantic citations from acyclic exact-byte commitments. Stable random WorkIDs support the first graph; release digests and predecessor links support the second.

The fourth requires an external witness such as an RFC 3161 timestamp authority (TSA), not a local clock or a mutable Git field. Conflating these layers produces systems that overclaim.

This paper presents ACSD (Anonymous Scholarly Claim and Disclosure), a deliberately narrow application profile. It uses standard content hashing, COSE signatures, and provenance concepts rather than proposing a new primitive. The profile is designed for content-anonymous release: the public paper and artifact do not name authors, but an account hosting them can still be linkable. It is therefore *not* described as author-unlinkable double blind publication.

The main contributions are:

1. a self-contained paper-release object with ordered anonymous author slots and per-author endorsements;
2. a two-layer construction that permits cyclic scholarly citations over stable WorkIDs while keeping content commitments acyclic;
3. signed UTF-8 byte-range citation witnesses and a restricted canonical JSON profile to reduce field-interpretation ambiguity;
4. explicit classifications for branches, equivocation, and copied content under a fresh lineage; and
5. an acyclic unanimous-approval target and closed capability algebra that prevent unsigned policy amplification and approval reuse after governance or PEC substitution; and
6. an executable prototype with a one-command release path, independent COSE verification, adversarial RFC 3161 checks, generated differential testing, and an axiom-free Lean model of the composition rules.

2 Scope and Threat Model

Assets. A release contains manuscript bytes, an ordered list of anonymous author keys, roles and contribution declarations, an AI-use declaration, and optional parent-release information. A series package contains references to completed releases, a series key, sequencing information, and cross-paper semantic relations. Public keys are pseudonymous identifiers.

Adversary. We consider a party that can copy public content, generate fresh keys, reorder or substitute unsigned metadata, present a signed but malformed JSON payload, and claim unsupported cross-paper citations. We also consider a series key that signs two incompatible objects. We do not

assume that a signature reveals a human identity, that a contribution declaration is factually true, or that a local timestamp is trusted.

Non-goals. ACS D does not prove authorship by a natural person, independent discovery, plagiarism, venue acceptance, peer review, or legal nonrepudiation. These are not merely unimplemented features: some require social, institutional, or legal evidence outside the cryptographic object. The profile instead preserves evidence that can be useful in a later dispute.

3 Release Layer

3.1 Canonical release object

Each *PaperRelease* has a random stable *WorkID*, a visible slot (*WorkID*, *version*, *line*), a content path, content length and digest, ordered author records, declared roles, contributions, an AI-use declaration, optional exact predecessor, and a list of citation witnesses. The release identifier is

$$\text{ReleaseID} = \text{urn:sha256:H}(\text{canonical}(R)).$$

The content digest is over exact manuscript bytes. The release object uses a deliberately restricted JSON encoding: recursively sorted ASCII object keys, no insignificant whitespace, and safe integers only. A verifier parses, re-canonicalizes, and compares bytes before accepting a signature. This is a profile rule, not a claim that it is a general JSON standard.

The inherited v1 series fixtures contain per-author release endorsements. The v2 command-line profile additionally constructs an `acsd-approval-target/v1` object containing the work id, sorted required key set, and the SHA-256 digests of the release, governance statement, and PEC. Each required key issues a tagged COSE Sign1 signature over the same canonical target. Verification recomputes every digest and also recomputes each key id from the supplied public key. Thus replacing a governance statement and PEC, even coherently, invalidates the target; recomputing the target cannot reuse the old signatures. A verifier can state that the pseudonymous keys jointly approved this exact tuple, but cannot infer who controls them without separate enrollment evidence.

3.2 Citation witnesses

The prototype recognizes only literal occurrences of a canonical token `urn:uuid:<work-id>` in the UTF-8 manuscript bytes. For every occurrence it records and signs

$$(\text{WorkID}, \text{byteOffset}, \text{byteLength}).$$

The published `reference_work_ids` list must equal the ordered de-duplicated projection of these witnesses. This prevents a series package from inventing a citation edge that has no corresponding token in its source release, and prevents a copied text from silently relabeling an old target as a new *WorkID*.

The rule intentionally does not parse PDFs, infer references from natural language, or determine whether a citation is intellectually appropriate. It is an exact byte-binding contract, not a semantic understanding claim.

For example, one UTF-8 source line contains the following token:

`urn:uuid:12345678-1234-1234-1234-123456789abc`

The witness records that WorkID, the byte offset of the first character of the token, and its byte length (45 bytes for this token). An editor that inserts a character before the token must regenerate the witness. This is intentionally mechanical: a source-level helper or build macro can insert the token, but v1 does not yet provide such editor integration, and hiding the token in a PDF would defeat this scanner.

A minimal release fragment is conceptually:

```
{ "slot": { "line": "main", "version": 1, "work_id": "urn:uuid:..." },
  "content": { "sha256": "..."},
  "reference_work_ids": [ "urn:uuid:..." ],
  "citation_witnesses": [ { "work_id": "urn:uuid:...",
    "byte_offset": 35, "byte_length": 45 } ] }
```

The ellipses above are presentation placeholders only. The released demo fixture binds a real manuscript's bytes, digest, and work identifier (the governance and PEC objects are real, not placeholders), using synthetic author identifiers. Real series releases are produced by the v1 fixture generator.

4 Series Layer and Cyclic Citations

4.1 Two graphs, two purposes

A paper remains verifiable after all series packages are removed. The optional series layer records relationships among independently complete releases. A signed series package contains

$$B_k = \text{Sign}_S(\text{seriesId}, \text{epoch}, \text{sequence}, \text{previousBundleHash}, \\ \text{rotationHash}, \text{rank}, \text{releases}, \text{heads}, \text{semanticEdges}).$$

The semantic graph has vertices WorkIDs and may contain $P \rightarrow Q$ and $Q \rightarrow P$. Its edges express declared citation or related-work relations. The commitment graph has series anchors and completed ReleaseIDs. It does not make a release depend on a future series package. Thus two authors can cite stable random WorkIDs before both final documents are content-addressed; a later package resolves each WorkID to a completed ReleaseID. The semantic graph may be cyclic while the commitment graph remains a DAG.

The package includes a signed structural rank strictly above every release version it resolves. Combining anchor-to-release edges with exact release-to-parent edges decreases this rank along a commitment path. This rank is a graph certificate, not a timestamp, publication date, or priority claim.

The package fields have narrow roles: *epoch* names the active series key generation; *rotationHash* commits its authorized key transition; *heads* maps each WorkID to its advertised line head; and *semanticEdges* records declared citation or related-work relations. The rank is only a finite measure certifying that the commitment graph has no cycle.

4.2 Revisions, branches, and conflicts

A revision names one exact predecessor. Multiple children of a parent are allowed, because a main manuscript and a computational appendix can be legitimate independent lines. A paper-level equivocation is instead detected when two valid, different releases occupy the same exclusive slot. A series-level equivocation is detected when two valid packages share the same series, epoch, sequence, and predecessor hash while committing to different contents.

An adversary can copy the content bytes, issue a fresh `WorkID` and keys, and obtain a valid new lineage. ACS D classifies this as content reuse under a different lineage, not as a cryptographic proof of plagiarism. That restraint is important: reuse may be authorized, collaborative, or independently derived.

As a concrete attack, suppose an attacker copies P-v1, changes its `WorkID`, and asks the series package to claim that it cites Q. The release verifier recomputes the content digest and byte witnesses; if the copied source does not contain the declared Q token, the package is rejected as an unsupported edge. If the attacker copies the token too, the package can still be valid under a new lineage. The evidence then says only “content reuse under a different lineage”; deciding plagiarism or authorization remains outside the verifier.

5 Time Evidence

Time is intentionally external to author signing. An RFC 3161 request contains the SHA-256 imprint of the frozen approval target. A response supports the limited statement that the TSA saw that imprint no later than its recorded `genTime` only when the verifier supplies a TSA signer-certificate or fingerprint pin independently of the package. A certificate and its fingerprint copied into the same mutable package are not a trust anchor. The adapter checks the request/response nonce, SHA-256 imprint, `id-ct-TSTInfo` content type, single signer identifier, critical and exclusive timeStamping EKU, ESS certificate identifier, signed attributes, and CMS signature. It implements an exact-signer pin, not general PKIX path construction or revocation. Its local test TSA is a protocol test and can never produce `EXTERNALLY_NOT_AFTER`.

There is no circularity: the timestamp receipt is a sidecar over the already signed target. A later revision can bind the receipt hash if a new signed object must refer to it. A timestamp does not prove creation at that moment, authorship, originality, or global first discovery. Multiple independently governed services can reduce single-service reliance, but their composition requires a separate policy.

6 Implementation and Evaluation

The v1 prototype uses Python for deterministic fixture generation and a zero-dependency Node.js verifier for strict CBOR/COSE and profile validation; the two paths are intentionally separate. The package is designed for offline replay: locked dependencies are included, and the verification driver regenerates fixtures, verifies a standalone release, independently verifies the series profile, builds a manifest, and verifies that manifest.

A command-line tool (`acsd.py`) turns the profile into an executable workflow: `keygen`, `init`, `approve`, `finalize`, `verify`, and a one-command `release` path for one or more locally available author keys. Approval status is derived from verified COSE files, never trusted from mutable state metadata. COSE uses the registered EdDSA value `-8`, requires tag 18, and rejects non-minimal or ambiguous CBOR encodings. A zero-dependency Node path independently verifies Python-produced approvals. The canonical JSON layer is exercised by 64 fixed Python–Node vectors plus 1,000 seeded generated cases; all generated cases agree on canonical bytes and text-layer verdicts. Three known object-layer float typing differences do not change the text-layer rejection result and remain recorded in the canonical specification.

The negative corpus includes the inherited rank, unsupported-edge, citation-offset, canonical-JSON, and path attacks. New cross-layer regressions replace a public key under an old key id; coherently replace governance and PEC; recompute an unsigned target and attempt to reuse old

Table 1: Executable evaluation corpus. The first seven rows are inherited v1 fixture results; the remaining rows are new v2 results reproducible from this release.

Object or check	Count / result
Standalone release versions	8 / 8 valid without series packages
Author endorsements	18 / 18 bound to the intended release
Signed series objects	7
Closed normal, attack, and indeterminate scenarios	13 / 13 expected outcomes
Independent profile checks	30 / 30 passed
Canonical JSON regression cases	11 / 11 Python–Node agreement
Public-manifest entries	113 / 113 verified
v2 differential canonical-JSON vectors	64 / 64 accept/reject agreement
Offline v2 unit-test methods	51 passed; 1 network test skipped
Seeded generated canonical cases	1000 / 1000 byte and verdict agreement
Independent approval implementations	Python producer / Node verifier
Real RFC 3161 interoperability	1 freeTSA response verified
New PEC theorems in the abstract core	16; no sorry/admit

approvals; alter COSE tags, algorithm identifiers, map keys, and lengths; replace a timestamp and rewrite the transport manifest; mismatch the nonce, signer, ECU, TSTInfo content type, or signature algorithm; and attempt to promote the local test TSA. All are rejected at their intended boundary.

The measurements remain prototype-scale rather than a throughput claim. In one local Windows run, five repetitions of an in-memory workload that canonicalizes objects, constructs and verifies SHA-256 manifest entries, and scans a linear rank certificate had median times of 2.23, 26.45, 227.54, and 718.81 ms for 10, 100, 1,000, and 5,000 objects respectively; peak traced memory at 5,000 objects was 7.04 MiB. This excludes filesystem, COSE, and network cost, so it only checks that the reference core has no immediate superlinear surprise.

The Lean development uses concrete approval targets, approvals, digest-linked events, a closed outcome type, and decomposed time evidence. Its 16 theorems prove, among other facts, that an approval for one target digest cannot be reused for a different target; a granted result implies acceptance and policy permission; key assent requires all target approvals; and external time requires exact-subject, signer-pin, nonce, profile, and non-local predicates. The model deliberately accepts signature validity and digest equality as inputs. It does not prove parser refinement, SHA-256 collision resistance, Ed25519 unforgeability, TSA honesty, or general PKIX correctness.

7 Related Work

The profile builds on established foundations. PROV-DM supplies a general vocabulary for entities, activities, agents, derivations, and bundles [1]. Software Heritage identifiers provide intrinsic identifiers for software objects [2]; Trusty URIs provide hash-bearing immutable artifact identifiers [3]. COSE defines the signed binary envelope used for endorsements [4]. SCITT specifies a broader architecture for signed statements and transparency receipts [5]. RFC 3161 specifies the timestamp protocol [6].

ACSD does not claim to replace these systems or invent their primitives. Its narrow contribution is an executable scholarly profile that combines: self-contained anonymous paper releases; a

Table 2: Capability boundary (not a performance comparison).

System	Primary capability	ACSD boundary
OpenTimestamps	Blockchain-anchored timestamp proofs	Optional time sidecar; no author-role or citation semantics
Software Heritage / SWHID	Persistent intrinsic identifiers for archived software objects	ACSD binds scholarly release roles, declarations and paper relations
Zenodo	Versioned archival record and DOI	ACSD’s byte-level policy and pseudonymous team assent remain explicit
RFC 3161	TSA-signed message-imprint time evidence	Exact not-after evidence only; never authorship or originality
ACSD	Release, governance, revision, witness and claim-policy profile	Composes these boundaries while retaining backend trust assumptions

separate series layer for mutual citations; exact text-token witnesses; explicit conflict predicates; and counterexamples against common overclaims about signatures, local clocks, and copied content.

8 Limitations and Future Work

Several extensions are intentionally deferred. First, relations across independently administered series require policy and an interoperable cross-series package model. Second, anti-equivocation evidence from transparency logs or gossip requires deployed service assumptions. Third, the parser, canonicalization, hash, and signature implementations have not been refined into the Lean model. Fourth, generated private keys are unencrypted PEM files; a production CLI needs encrypted or operating-system-backed storage. Fifth, exact TSA signer pinning does not replace PKIX path construction and revocation when a deployment requires those services. Finally, content-anonymous publication does not hide repository-account, network, timing, or public-key linkage, and richer citation/editor integration remains future work.

These are reasons to state the present result narrowly, not reasons to weaken the useful evidence it already preserves.

9 Conclusion

ACSD turns a vague “anonymous signed paper” idea into a limited but executable evidence profile. It binds exact released objects to pseudonymous-key approvals, supports ordered roles and revisions, permits mutual scholarly citation without cyclic hash dependencies, and rejects several malformed or misleading artifacts. It leaves human identity, originality, and independent time as claims requiring additional evidence. This division of claims is the central design principle: preserve what cryptography can verify, and make the remaining social questions explicit.

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